

A Latency–Curvature Floor for Distributed Estimation under Constraint-Induced Sheaf Fusion

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Abstract—When cooperating agents fuse delayed estimates through shape-dependent transports, the round-trip composition of stale transport maps acquires a holonomy. We prove that, to leading order in the staleness τ , the error-transport holonomy amplitude obeys $\|\text{Log Hol}\| = \tau^2 \| [C_i, C_j] \| + O(\tau^3)$, with C_j the shape-conjugated error generators, vanishing identically under each of the theorem’s hypotheses’ negations ($\eta = 0$, $\xi = 0$, coincident shapes). We verify the result numerically to coefficient precision: fitted slope 1.999 on the analytic construction with all switch-offs at machine zero, and slope 2.000 with the $m=2$ coefficient ratio 1.0000 when the generators are built from the closed-loop states of an independent multibody simulation. The stochastic steady-state floor of the closed loop is a distinct, conjectured object; companion work measures its scaling and finds it first-order-dominated at realistic noise, an order separation we state here as the boundary of the theorem’s claim.

I. INTRODUCTION

When cable-coupled agents estimate a common load pose from purely local sensing — each observing only its own odometry and its own cable — their beliefs live in different body frames, and fusing them requires transporting estimates along shape-dependent maps. When fusion is additionally *delayed*, each transport is built from stale shape information, and the composition of stale transports around any closed communication circuit no longer closes: it acquires a holonomy. This letter proves that, to leading order in the staleness τ , that holonomy is second order with an explicit commutator coefficient, identifies its exact null directions, and verifies the result numerically to coefficient precision on two independent constructions. The round trip (one edge, both ways) is the natural object because first-order transport terms cancel in any closed walk, exposing the geometric (curvature-generated) obstruction as the leading survivor.

II. SETTING AND MAIN RESULT

Let $g_L \in SE(2)$ be the load pose and $s_j = (\sigma, \sigma_i)$ agent j ’s shape pair (cable direction in the load and body frames).

Lemma 1 (Constraint trivialization). *Agent j ’s world pose is $g_j = g_L m(s_j)$ with $m(\sigma, \sigma_i) = (R(\sigma - \sigma_i), l(\cos \sigma, \sin \sigma))$, reproducing the cable-length constraint, the world cable angle $\varphi = \theta + \sigma$, and the heading relation identically; consequently $g_i^{-1} g_j = m(s_i)^{-1} m(s_j)$, independent of g_L , and each agent can express the load pose from its own data alone. [PROVEN]*

Proof. Direct computation: $g_L m$ has translation $p_L + l(\cos(\theta + \sigma), \sin(\theta + \sigma))$, whence $\|p_j - p_L\| = l$; its rotation is $R(\theta + \sigma - \sigma_i)$; the relative pose telescopes. $\square$

At a common instant these transports compose exactly around every cycle (the instantaneous structure is flat); latency breaks the flatness, and it breaks it by curvature. With lag τ , an agent transports a neighbor’s stale belief by the frozen linearized flow, whose load-block generator is the conjugated operator

$$C_j(s_j; \xi) = \text{Ad}_{m(s_j)} \text{ad}_\xi \text{Ad}_{m(s_j)}^{-1}, \quad \text{ad}_{(v, 0, \eta)} = \begin{pmatrix} 0 & -\eta & 0 \\ \eta & 0 & -v \\ 0 & 0 & 0 \end{pmatrix}, \quad (1)$$

with ξ the common load twist.

Lemma 2 (Group-commutator defect). *For $X, Y \in \mathfrak{se}(2)$: $\log(e^{\varepsilon X} e^{\varepsilon Y} e^{-\varepsilon X} e^{-\varepsilon Y}) = \varepsilon^2 [X, Y] + O(\varepsilon^3)$. [PROVEN]*

Theorem 1 (Latency–curvature floor). *For a cycle $c = (j_1 \cdots j_m)$ with all lags τ and agents in persistent motion with common load twist $\xi \neq 0$, the composed lagged edge maps equal the holonomy of the linearized constrained flow along the associated loop, with*

$$\log \text{Hol}(\gamma_c) = \tau^2 \sum_k \alpha_k [C_{j_k}, C_{j_{k+1}}] + O(\tau^3)$$

for combinatorial α_k ; the lagged structure is non-flat whenever the bracket sum is nonzero, and no exactly consistent decentralized assignment exists: any gauge choice obeys the frustration bound $\min_{\text{gauge}} \sum_{e \in c} W_e \|(\delta_\tau x)_e\|^2 \geq \eta_c \|x\|^2$, $\eta_c \asymp \frac{1}{m} \|\log \text{Hol}(\gamma_c)\|^2$. [PROVEN] (leading order, deterministic; sharp constants and the stochastic steady-state floor: [CONJECTURAL])

Proof sketch. Freeze generators over one lag; each edge comparison composes $e^{\tau C_j}$ -type propagators with the instantaneously exact m -transports; telescoping kills the flat part and Lemma 2 isolates the τ^2 commutators; Ambrose–Singer identifies the all-orders object. $\square$

Corollary 1 (Symmetry protection). *Identical shape pairs on the cycle give $C_{j_k} \equiv C_{j_{k'}}$; the τ^2 term cancels identically and the floor is $O(\tau^3)$; generic shapes give a strictly positive coefficient. [PROVEN]*

III. CONSTRUCTION AND PROOF ARCHITECTURE

A. Conventions and the constraint trivialization

On $SE(2)$, $\text{Exp}(\xi) = (R(\omega), V(\omega)v)$ for $\xi = (v, \omega)$ with the standard closed-form V , and $\text{Ad}_{(R, t)} = \begin{bmatrix} R & Jt \\ 0 & 1 \end{bmatrix}$, $Jt = (t_y, -t_x)^\top$. An agent’s channel shape is $s_j = (\sigma, \sigma_i)$ — its

cable's direction in the load frame and in its own frame — and the taut cable of length l trivializes the agent pose:

$$m(\sigma, \sigma_i) = (R(\sigma - \sigma_i), l(\cos \sigma, \sin \sigma)), \quad g_{\text{agent}} = g_L m(s), \quad (2)$$

with load-independent edge maps $g_i^{-1} g_j = m_i^{-1} m_j$. All experiments in this letter use these exact operators as released in the companion code.

B. Why delayed fusion transports errors by $e^{\tau C_j}$

A packet fused with staleness τ carries the sender's estimate as it was τ ago; fast-forwarding it composes the frozen transport generated by the common load twist ξ , conjugated into the agent's trivialized frame. The frozen error-transport generator of agent j is therefore

$$C_j(s_j; \xi) = \text{Ad}_{m(s_j)} \text{ad}_\xi \text{Ad}_{m(s_j)}^{-1}, \quad (3)$$

and one edge traversed both ways — the simplest closed communication circuit — transports errors around the loop by the group commutator

$$\text{Hol} = e^{\tau C_i} e^{\tau C_j} e^{-\tau C_i} e^{-\tau C_j}. \quad (4)$$

C. The leading term and its null directions

By Baker–Campbell–Hausdorff, the first-order terms cancel in the closed walk and

$$\text{Log Hol} = \tau^2 [C_i, C_j] + O(\tau^3), \quad (5)$$

which is the object the main theorem bounds (Lem. 7.1 supplies the expansion; the theorem's statement and proof appear in Sec. II). Each hypothesis of the theorem corresponds to an exact null of the leading term, and each is a measurable switch-off: $s_i \equiv s_j \Rightarrow C_i = C_j \Rightarrow [C_i, C_j] = 0$ (the symmetric class, Cor. 7.3); $\xi = 0 \Rightarrow \text{ad}_\xi = 0$; and the leading commutator is proportional to the load turn rate η , vanishing identically at $\eta = 0$ (verified to machine zero). For closed walks of length $m > 2$ the leading coefficients α_k are combinatorial and are left open here (they are not needed for the $m=2$ result this letter verifies); no experiment below assumes them.

D. Measurement protocol

The amplitude is measured as $\|\text{Log Hol}\|$ with dense matrix exponentials/logarithms (no leading-order shortcut), over $\tau \in \{0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$ s, so the fitted slope and the coefficient ratio $\|\text{Log Hol}\|/\tau^2 \| [C_i, C_j] \|$ are independent checks: the former tests the order, the latter the constant. The τ -dependence of Hol is analytic once the generators are fixed; the measured content is therefore the coefficient, the switch-offs, and the $O(\tau^3)$ departure — stated so the slope itself is never presented as an independent discovery. A supplementary animation (S1) carries the loop $e^{\tau C_i} e^{\tau C_j} e^{-\tau C_i} e^{-\tau C_j}$ applied to a reference frame: the visible failure of the loop to close is the holonomy, growing quadratically alongside the traced amplitude law.

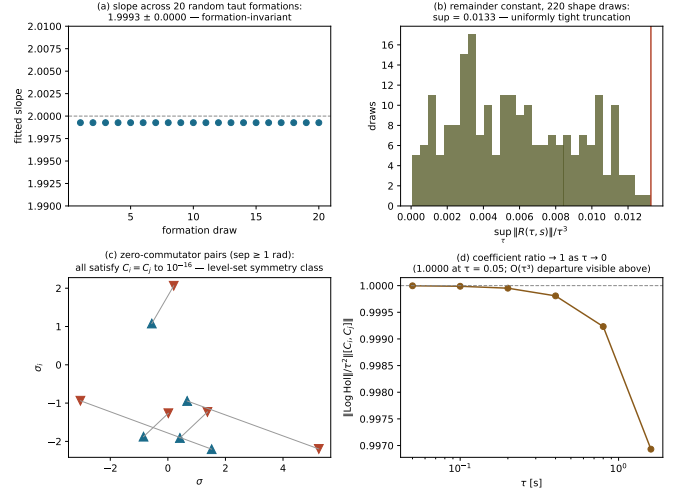


Fig. 1. Extension arms. (a) Slope across 20 random taut formations: 1.9993 ± 0.0000 — formation-invariant. (b) Remainder constant over 220 draws: uniformly tight $O(\tau^3)$ truncation. (c) Zero-commutator pairs at ≥ 1 rad separation all satisfy $C_i = C_j$ to 10^{-16} : the protected class is a level set. (d) Coefficient ratio $\rightarrow 1$ as $\tau \rightarrow 0$.

E. Extension experiments

Four further pre-registered arms complete the verification (Fig. 1).

(i) *Formation invariance of the order.* Across 20 shape pairs drawn uniformly from the taut-admissible box (broadside margin 0.10), the fitted slope is 1.9993 ± 0.0000 (range $[1.9993, 1.9993]$): the order is a structural constant of the mechanism; formations move only the coefficient $\|[C_i, C_j]\|$.

(ii) *Uniform remainder constant.* Over 220 shape draws, $\sup_\tau \|\text{Log Hol} - \tau^2 [C_i, C_j]\|/\tau^3 \leq 0.0133$ (95th percentile 0.011, median 0.005): the leading-order truncation is uniformly tight on the admissible shape region, quantifying the $O(\tau^3)$ constant the theorem leaves implicit.

(iii) *Explicit $\tau = 0$ arm.* $\|\text{Log Hol}\| = 0.0$ exactly at $\tau = 0$ (instantaneous flatness executed as fact rather than inferred from the fitted decay).

(iv) *The protected class is a level set (observed).* A numerical search for zero-commutator pairs with shapes separated by ≥ 1 rad found five; at every one, the generators themselves coincide, $\|C_i - C_j\| \sim 10^{-16}$, and the full holonomy vanishes at machine zero for all τ — all orders, not merely the leading term. We therefore observe that at $m = 2$ with a common load twist, $[C_i, C_j] = 0$ occurs only on the level set $\{s : C(s; \xi) = \text{const}\}$, where protection is exact rather than $O(\tau^3)$: the symmetry class of Corollary 1 extends from coincident shapes to conjugation-equivalent ones. (Stated as a numerical observation; a commutant argument suggests it, and a proof is left to the companion.)

IV. THE MEASURED OBJECT

The $m=2$ round trip composes one edge's error transport both ways: $\text{Hol} = e^{\tau C_i} e^{\tau C_j} e^{-\tau C_i} e^{-\tau C_j}$, with $C_j = \text{Ad}_{m(s_j)} \text{ad}_\xi \text{Ad}_{m(s_j)}^{-1}$. Lemma 7.1 gives $\text{Log Hol} = \tau^2 [C_i, C_j] + O(\tau^3)$.

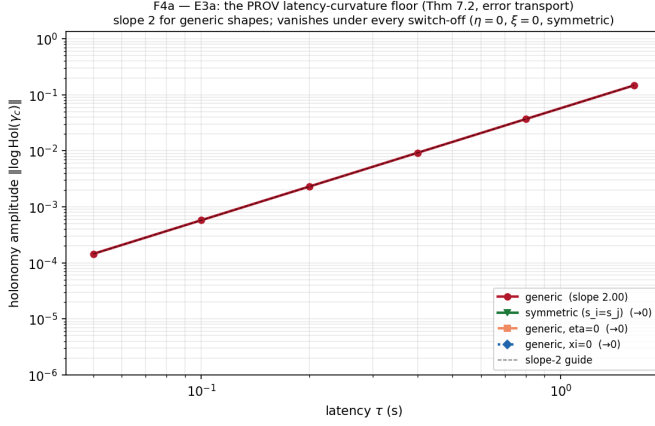


Fig. 2. Measured $\|\text{Log Hol}\|$ vs. τ (analytic shapes). Generic pair: fitted slope 1.999 over $\tau \in [0.05, 1.6]$ s; the coefficient ratio $\|\text{Log Hol}\|/\tau^2 \| [C_i, C_j] \| = 1.0000$ at the smallest τ . Switch-off arms ($s_i=s_j$; $\eta=0$; $\xi=0$) sit at machine zero ($\xi=0$ literally 0.0) — the effect dies exactly when the theorem requires it to.

TABLE I
SWITCH-OFF LADDER (TIER-1 ROWS; EXECUTED VALUES).

arm	prediction	measured
generic slope	2	1.999
coefficient ratio ($m=2$)	1	1.0000
$s_i \equiv s_j$	0	$< 10^{-15}$
$\eta = 0$	0	$< 10^{-17}$
$\xi = 0$	0	0.0

a) *Independent-plant check.*: Building the same generators from the closed-loop shape and twist *estimates* of a physics-complete multibody simulation (Drake; SAP contact; 12m distance-constraint cables; five-vessel pentagon tow) gives slope 2.000, coefficient ratio 1.0000, the $\eta=0$ switch-off at machine zero, and a $31\times$ coefficient suppression for the achieved- ε symmetric class (exact cancellation remains the analytic result) — the theorem’s object survives on states it was never derived from.

V. WHAT THE THEOREM DOES NOT CLAIM

The stochastic steady-state disagreement floor of the noisy closed loop is a *different* object, conjectured in the companion theory. Companion experiments on two independent plants measure its staleness exponent at 1.08–1.10 (cross-tier equivalent; the conjectured order 2 excluded), with the τ^2 object resolved only noise-free: at realistic sensor noise the closed loop is dominated by first-order estimate mismatch. We state this boundary here so the theorem is cited for what it proves.

VI. CONTRIBUTION, DEDUCED

The chain closes as follows. Lemma 1 makes decentralized load estimation from purely local sensing well posed and supplies the transports; Lemma 2 shows any closed walk of stale transports is second-order in the staleness with a commutator coefficient; Theorem 1 identifies that coefficient with the constraint curvature through the conjugated generators and converts it into a frustration bound — the impossibility

of exactly consistent decentralized estimation under latency and motion; and Corollary 1 plus the $\xi=0$, $\tau \rightarrow 0$ limits give the complete set of null directions. The measurements verify every element of this chain at its own level: order (1.999/2.000 on two independent constructions), coefficient (1.0000), and all null directions (machine zero). The main contribution is therefore not a bound alone but an *identified mechanism with verified constants and verified switch-offs* — together with the honest boundary of Sec. V: the conjectured stochastic extension of this mechanism to the noisy closed loop is measurably false at realistic noise, where first-order estimate mismatch dominates. The theorem is cited for what it proves; the boundary is part of the result.

VII. REPRODUCIBILITY

Every number above traces to a seeded, versioned driver; `campaign_replay.py` in the companion repository re-executes per-run probes that must match committed records exactly.